CMSys Documents — System Architecture & Technical Specification

Document Reference: `SLN/CCED/ENG/SAD-2026/01`
Classification: `RESTRICTED / OFFICIAL USE ONLY`
Organization: Sri Lanka Navy • Eastern Naval Command • Command Civil Engineering Department (East) [CCED(E)]
Station: SLN Dockyard, Trincomalee
System Version: Release 2.4.0 (Enterprise Gold)

📄 Complete Master Documentation in PDF

The comprehensive **37-Page Full System Architecture and Specification Document** has been compiled and rendered to PDF:

- **Local File System (Workspace Root):**
  (1.67 MB)
- **Archival Docs Directory:**
  (1.67 MB)
- **Direct Web Access via Localhost:**
  http://localhost/CMSys Documents/CMSys_Documents_System_Architecture_and_Specification.pdf

🏛️ High-Level System Architecture Overview

🎯 Core Architectural Capabilities

1. 100% Compliance with Naval Book of Reference 7 (BR 7)

- **Precedence Levels:** `ROUTINE`, `PRIORITY`, `IMMEDIATE`, `FLASH`

- **Security Classifications:** `UNCLASSIFIED`, `RESTRICTED`, `CONFIDENTIAL`, `SECRET`
- **Addressing Matrix:** Formal Functional Appointment (`To:`), Hierarchical Command Routing (`Through:`), and Designated Action Officer (`Attention:`)
- **Dynamic Tone Engine:** Automated linguistic switching between `junior` ("*I have the honour to submit...*" / "*I remain, Sir, Your obedient Servant*"), `senior` ("*You are requested to...*"), and `equal` ("*I have the honour to refer to...*")
- **Hierarchical Paragraphing:** Arabic numerals (`1.`), alphabetical sub-paragraphs (`(a)`), and roman sub-sub-paragraphs (`(i)`)
- **Direct Registry Linkage:** Single-click conversion from draft letters to registered outgoing records in `document_registry`

2. Bilingual Minute Sheet Studio (Sinhala & English)

- **M01 & M02 Approval Blocks:** Dual-minute layout with Originator submission (M01) and Departmental Endorsement (M02)
- **Automated Rank Transliteration:** Converts English naval abbreviations (`Commander (CE)`, `MCPO`, `Leading`, etc.) into approved military Sinhala designations (`කොමදෝරු (සිවිල්)`, `ප්‍රධාන මාණ්ඩලික සුළු නිලධාරී`, `නායක නාවික`)
- **Micro-Typographic Controls:** Real-time calibration of divider width (50%–100%), right margin offset (10mm–35mm), font size (8.5pt–13pt), and leading (1.3–2.2)

3. Atomic Sequential Reference Engine

- **Tokenized Patterns:** Evaluates `{YYYY}`, `{YY}`, `{MM}`, and `{SEQ}` with zero-padding (e.g., `CCED/NHQ/2026/0012` and `CCED/A/10/2216/26`)
- **Concurrency Protection:** Database-level transactional locking (`SELECT ... FOR UPDATE`) prevents duplicate reference issuance across concurrent workstation dispatches

4. Hybrid Dual-Storage & Cloud Resilience

- **Local-First Durability:** Zero dependency on external networks for core operations
- **Asynchronous Cloud Sync:** Automatic background replication to Firebase Realtime Database
- **Latency Shielding:** Strict 3-second cURL timeouts ensure UI never freezes during cloud or satellite dropouts

5. Zero-Build Frontend Deployment

- Pure vanilla ECMAScript 6+ modules with pre-compiled Tailwind CSS v3
- Zero NPM packages, Node runtime, or bundler build steps needed in production

- Embedded local TrueType fonts (`Abhaya Libre` , `Gemunu Libre` , `Noto Serif Sinhala` , `Yaldevi` , `Stick No Bills` , `Google Sans` , `Humanist 521`) for total air-gapped security

📊 Relational Database Schema Summary (8 Tables)

Table Name	Storage Engine	Records / Purpose	Key Fields
`document_registry`	InnoDB	Inward & Outward master ledger	`id` , `ref_no` , `office_section` , `flow type` , `position_trail` , `date_recorded`
`registry_prefixes`	InnoDB	Departmental sequence numbering rules	`id` , `office_section` , `category` , `prefix pattern` , `next_sequence` , `padding_length`
`file_registry`	InnoDB	Physical folders, box files & folios	`id` , `file_no` , `subject` , `cabinet_location` , `file_box_ref` , `current_volume` , `total_folios`
`naval_letters`	InnoDB	BR 7 naval correspondence records	`id` , `letter_ref` , `security_class` , `precedence` , `br7_tone` , `paragraphs` , `signatory_name`
`naval letter_templates`	InnoDB	Reusable SOP military letter templates	`id` , `title` , `category` , `security_class` , `br7_tone` , `paragraphs` , `copy_to_list`
`minute_templates`	InnoDB	Bilingual minute sheet templates	`id` , `lang` , `title` , `paragraphs` , `recommendation` , `m02_enabled` , `sign_title`
`job_minutes`	InnoDB	Civil engineering job minutes	`id` , `ref_no` , `end_user` , `subject` , `project id` , `linked_estimate_id` , `position_trail`
`system document_settings`	InnoDB	Centralized system typography & presets	`setting key` , `setting_value` , `updated_at`

🔗 Quick Access Links

- **Open Web Application:** http://localhost/CMSys Documents/
- **Open Complete System Document (PDF):**
- **Open Document Source (HTML):**

- **Database Setup & Migrations:**
- **Central API Gateway:**